



Circumference and diameter

THE DISTANCE AROUND THE EDGE OF A CIRCLE IS CALLED THE CIRCUMFERENCE; THE DISTANCE ACROSS THE MIDDLE IS THE DIAMETER.

All circles are similar because they have exactly the same shape. This means that all their measurements, including the circumference and the diameter, are in proportion to each other.

The number pi

The ratio between the circumference and diameter of a circle is a number called pi, which is written π . This number is used in many of the formulas associated with circles, including the formulas for the circumference and diameter.

symbol for pi

$$\pi \approx 3.14$$

value to 2 decimal places

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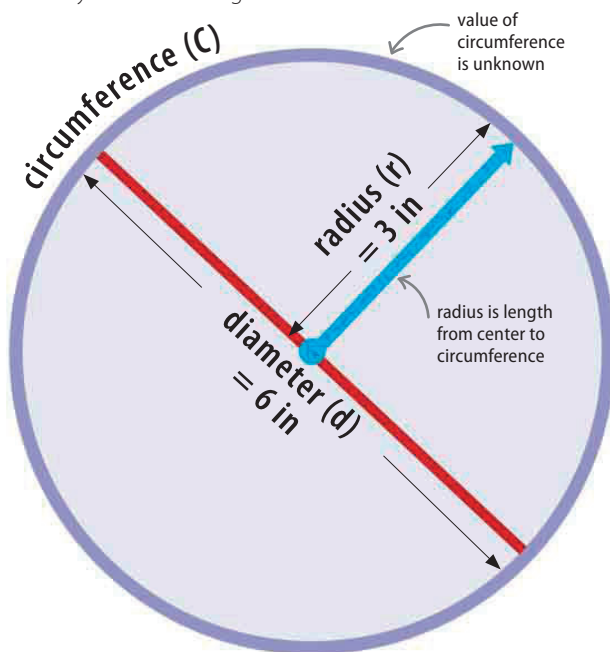
Area of a circle 142–143 ▶

◁ The value of pi

The numbers after the decimal point in pi go on for ever and in an unpredictable way. It starts 3.1415926 but is usually given to two decimal places.

Circumference (C)

The circumference is the distance around the edge of a circle. A circle's circumference can be found using the diameter or radius and the number pi. The diameter is always twice the length of the radius.



circumference

π is a constant

radius

$$C = 2\pi r$$

circumference

π is a constant

diameter

$$C = \pi d$$

◁ Formulas

There are two circumference formulas. One uses diameter and the other uses radius.

The formula for circumference shows that the circumference is equal to pi multiplied by the diameter of the circle.

$C = \pi d$

d is the same as $2 \times r$, the formula can also be written $C = 2\pi r$

Substitute known
values into the formula for circumference. Here, the radius of the circle is known to be 3 in.

$C = 3.14 \times 6$

pi is 3.14 to two decimal places

Multiply the numbers
to find the length of the circumference. Round the answer to a suitable number of decimal places.

$C = 18.8 \text{ in}$

18.84 is rounded to one decimal place

△ Finding the circumference

The length of a circle's circumference can be found if the length of the diameter is known, in this example the diameter is 6 in long.